

364. Total number of students = 54×30 .

$$\therefore \text{Required number of rows} = \left(\frac{54 \times 30}{45} \right) = 36.$$

365. Clearly, we have:

$$x + 1 = y \Leftrightarrow x - y = -1 \quad \dots(i)$$

$$\text{And, } 2(x - 1) = y \Leftrightarrow 2x - y = 2 \quad \dots(ii)$$

Subtracting (i) from (ii), we get: $x = 3$. Putting $x = 3$ in (i), we get: $y = 4$.

366. Let the number of benches in the class be x .

$$\text{Then, } 6(x + 1) = 7x - 5 \Leftrightarrow x = 11.$$

Hence, number of students in the class

$$= 6(x + 1) = 6 \times 12 = 72.$$

367. Let the number of benches in the class be x .

$$\text{Then, } 4(x - 3) = 3x + 3 \Leftrightarrow x = 15.$$

$$\therefore \text{Total number of students} = 3 \times 15 + 3 = 48.$$

368. Let the number of students in a row be x and the total number of rows be y . Then,

$$\text{total number of students in the class} = xy.$$

$$\therefore (x + 4)(y - 2) = xy \Rightarrow xy - 2x + 4y - 8 = xy$$

$$\Rightarrow 2x - 4y = -8 \Rightarrow x - 2y = -4 \quad \dots(i)$$

$$\text{And, } (x - 4)(y + 4) = xy \Rightarrow xy + 4x - 4y - 16 = xy$$

$$\Rightarrow 4x - 4y = 16 \Rightarrow x - y = 4 \quad \dots(ii)$$

Subtracting (i) from (ii), we get: $y = 8$.

Putting $y = 8$ in (ii), we get: $x = 12$.

$$\text{Hence, total number of students in the class} = xy = (12 \times 8) = 96.$$

369. Let Shyam's share be ₹ x . Then, Ram's share

$$= ₹ \left(x + \frac{3}{4}x \right) = ₹ \frac{7x}{4}.$$

$$\therefore x + \frac{7x}{4} = 275 \Leftrightarrow \frac{11x}{4} = 275 \Leftrightarrow x = \frac{275 \times 4}{11} = 100.$$

$$\text{So, Ram's share} = ₹ \left(\frac{7 \times 100}{4} \right) = ₹ 175.$$

370. Total number of sweets = $(64 \times 7 + 15) = 448 + 15 = 463$.

371. Let the whole property be worth ₹ x . Then,

$$\text{wife's share} = ₹ \frac{x}{2}; \text{ Son's share} = ₹ \left(\frac{1}{2} \text{ of } \frac{x}{2} \right) = ₹ \frac{x}{4}.$$

$$\text{Remaining share} = ₹ \left[x - \left(\frac{x}{2} + \frac{x}{4} \right) \right] = ₹ \frac{x}{4}.$$

$$\therefore \text{Share of each daughter} = ₹ \left(\frac{1}{2} \text{ of } \frac{x}{4} \right) = ₹ \frac{x}{8}.$$

372. Let the number of children be x . Then, $10x + 3 = 11x - 4 \Rightarrow x = 7$.

$$\therefore \text{Number of sweets} = (10 \times 7 + 3) = 73.$$

373. Let the number of boys be x . Then, number of girls = $(70 - x)$.

$$\therefore 2x + 3(70 - x) = 180 \Leftrightarrow x = 30.$$

374. Let the total number of cows in the herd be x .

$$\text{Then, share of fourth son} = x - \left(\frac{x}{2} + \frac{x}{4} + \frac{x}{5} \right) = x - \frac{19x}{20} = \frac{x}{20}.$$

$$\therefore \frac{x}{20} = 7 \text{ or } x = 140.$$

375. Let the total property be worth ₹ x .

$$\text{Then, elder son's share} = ₹ \left(\frac{5x}{11} \right). \text{ Balance} = ₹ \left(x - \frac{5x}{11} \right)$$

$$= ₹ \frac{6x}{11}.$$

$$\text{Younger son's share} = ₹ \left(\frac{5}{11} \text{ of } \frac{6x}{11} \right) = ₹ \left(\frac{30x}{121} \right).$$

$$\therefore \text{Widow's share} = ₹ \left[x - \left(\frac{5x}{11} + \frac{30x}{121} \right) \right]$$

$$= ₹ \left(x - \frac{85x}{121} \right) = ₹ \left(\frac{36x}{121} \right).$$

$$\frac{36x}{121} = 3600 \Leftrightarrow x = \frac{3600 \times 121}{36} = 12100.$$

$$\text{So, elder son's share} = ₹ \left(\frac{5}{11} \times 12100 \right) = ₹ 5500;$$

$$\text{Younger son's share} = ₹ \left(\frac{30}{121} \times 12100 \right) = ₹ 3000.$$

376. Suppose each boy gets ₹ x . Then, each woman gets ₹ $2x$ and each man gets ₹ $(x + 2x)$ i.e., ₹ $3x$.

$$\text{So, } 10 \times 3x + 12 \times 2x + 20x = 370 \Leftrightarrow 30x + 24x + 20x = 370 \Leftrightarrow 74x$$

$$= 370 \Leftrightarrow x = 5.$$

$$\therefore \text{Amount received by 10 men} = 30x = ₹ (30 \times 5) = ₹ 150.$$

377. Let D's share = ₹ x . Then, C's share = ₹ x .

$$\text{B's share} = ₹ (x + 125). \text{ A's share} = ₹ (x + x + 125) = ₹ (2x + 125).$$

$$\therefore (2x + 125) + (x + 125) + x + x = 750$$

$$\Leftrightarrow 5x = 500$$

$$\Leftrightarrow x = 100.$$

$$\text{Hence, A's share} = 2x + 125 = ₹ (2 \times 100 + 125) = ₹ 325.$$

378. Let Gagan's share = ₹ x .

$$\text{Then, Sachin's share} = ₹ \left(\frac{x}{5} \right) \text{ and Rohit's share} = ₹ \left(\frac{2x}{5} \right).$$

$$\therefore \frac{2x}{5} + \frac{x}{5} + x = 1000 \Leftrightarrow 8x = 5000 \Leftrightarrow x = 625.$$

379. If the second boy takes one-third of a certain number of marbles, the third boy is left with two-thirds of that number, which is twice the number with the second boy, regardless of the total number of marbles. Hence, the total number of marbles cannot be determined.

380. Wife's share = $\frac{1}{2}$. Remaining part = $\left(1 - \frac{1}{2} \right) = \frac{1}{2}$.

$$\text{Share of 3 sons} = \left(\frac{2}{3} \text{ of } \frac{1}{2} \right) = \frac{1}{3}. \text{ Remaining part}$$

$$= \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{1}{6}.$$

$$\text{Each daughter's share} = \frac{1}{4} \times \frac{1}{6} = \frac{1}{24}.$$

$$\text{Let the total money be ₹ } x. \text{ Then, } \frac{1}{24} x = 20000$$

$$\Leftrightarrow x = 20000 \times 24 = 480000.$$

$$\therefore \text{Each son's share} = ₹ \left[\frac{1}{3} \times \left(\frac{1}{3} \times 480000 \right) \right] = ₹ 53,333.33.$$